$$m \geq \delta^2 \left[1 - \frac{\log 2 + I(Z;J)}{\log M} \right]$$

$\tilde{\theta}$

$$I(Z;J) \leq \frac{1}{M^2} \sum_{i,j=1}^{M} D_{KL}\left(\mathbb{P}^n_{\theta^{(i)}} \,\|\, \mathbb{P}^n_{\theta^{(j)}} \right)$$

$$m \geq \delta^2 \left[1 - \frac{\log 2 + \cdots}{c\, s \log d/s} \right]$$

Obs. $\rightarrow$ $\underline{y} = \underbrace{A\theta^{(i)}}_{\text{const.}} + \underline{\epsilon}$ $\qquad \underline{\epsilon} \sim \mathcal{N}(\underline{0}, \sigma^2 I)$

$$\underline{y} - A\theta^{(i)} \stackrel{d}{=} \underline{\epsilon}$$

$$\underline{y} \sim \mathcal{N}(A\theta^{(i)}, \sigma^2 I)$$

$$D_{KL}\left(\mathbb{P}^n_{\theta^{(i)}} \,\|\, \mathbb{P}^n_{\theta^{(j)}} \right) = \frac{\| A\theta^{(i)} - A\theta^{(j)} \|^2}{2\sigma^2}$$

$$= \frac{\| A \overbrace{(\theta^{(i)} - \theta^{(j)})}^{\text{at most } 2s\text{-sparse}} \|^2}{2\sigma^2}$$

RIP

$\forall \theta^* : \|\theta^*\|_0 \leq 2s$

$$(1 - \gamma_{2s}) \|\theta^*\|_2^2 \leq \left\| \frac{A\theta^*}{\sqrt{n}} \right\|_2^2 \leq (1 + \gamma_{2s}) \|\theta^*\|_2^2$$

$$\Rightarrow D_{KL}\left(\mathbb{P}^n_{\theta^{(i)}} \,\|\, \mathbb{P}^n_{\theta^{(j)}} \right) \leq \frac{n(1+\gamma_{2s}) \boxed{\|\theta^{(i)} - \theta^{(j)}\|_2^2}}{2\sigma^2}$$

$$\leq 2\cancel{s} \times 8 \frac{\delta^2}{\cancel{s}} = 16\delta^2$$

$$\tilde{\theta}^{(i)} = 2\sqrt{2}\,\delta \frac{\theta^{(i)}}{\sqrt{s}}$$

$$\|\theta^{(i)} - \theta^{(j)}\|_2^2 \geq \cancel{8}^{4} \frac{\delta^2}{\cancel{s}} \left[\frac{\cancel{s}}{\cancel{2}} \right]$$

$$\Rightarrow \quad \| \theta^{(i)} - \theta^{(j)} \|_2 \geq 2\delta$$

$$m \geq \delta^2 \left[1 - \frac{\log 2 + 16n(1+\gamma_{2s})\delta^2 / 2\sigma^2}{c\, s \log d/s} \right]$$

it suffices to choose δ s.t. the ratio becomes $1/2$ $\rightarrow$ $\log 2 + 8n(1+\gamma_{2s})\delta^2/\sigma^2 = \frac{1}{2} c s \log d/s$

so that the 2^{nd} factor is pos.

$\Rightarrow$

$$\delta^2 = \sigma^2 \left(\tfrac{1}{2} c s \log d/s - \log 2 \right) / n(1+\gamma_{2s})$$

$$m \geq \frac{\sigma^2}{2} \left(\tfrac{1}{2} c s \log d/s - \log 2 \right) / [n(1+\gamma_{2s})]$$

$$= \omega \left(\frac{\sigma^2 s \log d/s}{n(1+\gamma_{2s})} \right). \qquad \square$$

Ch. 6 Covariance Estimation

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^{n} (\underline{x}_i - \hat{\underline{\mu}})(\underline{x}_i - \hat{\underline{\mu}})^T$$

$$\mathbb{P}\left(||| \hat{\Sigma} - \Sigma |||_2 \leq \delta \right) \overset{?}{\geq} 1 - e^{-cn\delta^2}$$

Theorem 6.1: If $X = \begin{bmatrix} \underline{x}_1 \\ \vdots \\ \underline{x}_n \end{bmatrix}$ comes from a (zero-centered) Σ-Gaussian ensemble, then for all $\delta > 0$

$$\mathbb{P}\left(\frac{\sigma_{max}(X)}{\sqrt{n}} \geq \gamma_{max}(\sqrt{\Sigma})(1+\delta) + \sqrt{\frac{tr(\Sigma)}{n}} \right) \leq e^{-n\delta^2/2}.$$

Example: $\Sigma = I$ $\quad \gamma_{max}(\sqrt{I}) = \gamma_{max}(I) = 1.$

$$\sqrt{\frac{tr(\Sigma)}{n}} = \sqrt{\frac{d}{n}}$$

$$\mathbb{P}\left(\frac{\sigma_{max}(X)}{\sqrt{n}} \geq (1+\delta) + \sqrt{d/n} \right) \leq e^{-n\delta^2/2}$$

$$\hat{\Sigma} = \frac{1}{n} X^T X \qquad \Sigma = I$$

$$|||\frac{1}{n} X^T X - I|||_2 = \sup_{\|v\|_2 = 1} \Big| \overbrace{v^T \left[\frac{1}{n} X^T X - I\right] v}^{1/n \|Xv\|_2^2 \; - \; \overbrace{\|v\|_2^2}^{1}} \Big|$$

$$= \sup_{\|v\|_2 = 1} \left\| \left(1/n \, X^T X - I\right) v \right\|_2$$